Requirements Gathering

Functional Requirements

1. User Authentication and Authorization

- **FR-1.1** The system needs secure login functionality for different user roles:
  - Logistics Manager
  - Warehouse Staff
  - Customer Service Representative
  - Driver/Carrier
  - Customer (public portal)
- **FR-1.2** The system shall implement role-based access control (RBAC) to restrict functionality based on user roles.
- **FR-1.3** The system shall maintain user session management with automatic timeout.

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2. Order and Shipment Management

- **FR-2.1** The system needs to allow warehouse staff to register new orders with the following info:
  - Destination address
  - Package weight and dimensions
  - Assigned carrier
  - Special handling instructions
  - Customer information
- **FR-2.2** The system shall provide order consolidation functionality to combine multiple orders into single shipments.
- **FR-2.3** The system shall automatically generate shipping documents.
- **FR-2.4** The system shall allow warehouse staff to update order status when carriers collect packages.
- **FR-2.5** The system shall maintain a centralized dashboard showing all active orders and their current status.

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3. Carrier Integration and Rate Management

- **FR-3.1** The system shall integrate with multiple third-party carriers (FedEx, UPS, DHL, local carriers).
- **FR-3.2** The system shall automatically retrieve and compare shipping rates from integrated carriers.
- **FR-3.3** The system shall display transit times for different carrier services.
- **FR-3.4** The system shall allow logistics managers to select optimal carriers based on cost and delivery time criteria.
- **FR-3.5** The system shall support carrier API integration for real-time rate updates.

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4. Real-time Tracking and Visibility

- **FR-4.1** The system shall provide real-time tracking updates through carrier API integration.
- **FR-4.2** The system shall allow customers to track orders using tracking numbers through a public portal.
- **FR-4.3** The system shall display current package location and estimated delivery dates.
- **FR-4.4** The system shall allow customer service representatives to access detailed tracking information for customer inquiries.
- **FR-4.5** The system shall maintain tracking history for all shipments.

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5. Route Optimization

- **FR-5.1** The system shall generate optimized delivery routes for drivers.
- **FR-5.2** The system shall consider factors such as:
  - Traffic conditions
  - Delivery time windows
  - Package priorities
  - Vehicle capacity constraints
- **FR-5.3** The system shall provide turn-by-turn navigation through a mobile application.

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6. Mobile Application for Drivers

- **FR-6.1** The system shall provide a mobile application for drivers with the following features:
  - Optimized route display
  - Package scanning capability
  - Proof of delivery capture
- **FR-6.2** The system shall allow drivers to capture proof of delivery through:
  - Delivery photographs
  - GPS location confirmation
- **FR-6.3** The system shall enable drivers to report delivery exceptions with predefined reason codes.

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7. Reporting and Analytics

- **FR-7.1** The system shall generate KPI dashboards for logistics managers showing:
  - On-time delivery rates
  - Shipping costs per order
  - Carrier performance metrics
  - Exception rates
- **FR-7.2** The system shall provide detailed performance reports including:
  - Cost analysis reports
  - Carrier performance comparisons
  - Delivery success rate analysis
  - Historical shipment data
- **FR-7.3** The system shall allow report export in multiple formats (PDF, Excel, CSV).
- **FR-7.4** The system shall support custom report generation with date range filters.

8. Notifications and Alerts

- **FR-8.1** The system shall send automated notifications for:
  - Package dispatch
  - Out for delivery status
  - Successful delivery
  - Delivery exceptions
  - Delayed shipments
- **FR-8.2** The system shall support multiple notification channels:
  - Email notifications
  - SMS notifications
  - In-system alerts

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9. Customer Portal

- **FR-9.1** The system shall provide a public-facing customer portal for order tracking.
- **FR-9.2** The system shall allow customers to:
  - Track orders using tracking numbers
  - View delivery history
  - Update delivery preferences
  - Request delivery notifications
- **FR-9.3** The system shall display estimated delivery windows and allow delivery scheduling where supported.

Non-Functional Requirements

1. Performance Requirements

- **NFR-1.1** The system shall respond to user queries within 3 seconds under normal load conditions.
- **NFR-1.2** The system shall process order registrations within 2 seconds.
- **NFR-1.3** The system shall update tracking information within 5 minutes of receiving carrier updates.
- **NFR-1.4** The mobile application shall load delivery manifests within 2 seconds.
- **NFR-1.5** The system shall generate reports within 10 seconds.

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2. Scalability Requirements

- **NFR-2.1** The system shall be horizontally scalable to handle growing transaction volumes.
- **NFR-2.2** The system shall scale to support additional carriers without architectural changes.

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3. Availability Requirements

- **NFR-3.1** The system shall implement automatic failover mechanisms to ensure service continuity.

- **NFR-3.2** The mobile application shall function offline for basic operations (viewing manifests, POD capture).

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4. Reliability Requirements

- **NFR-4.1** The system shall maintain data consistency across all modules.
- **NFR-4.2** The system shall handle carrier API failures gracefully with retry mechanisms.

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5. Usability Requirements

- **NFR-5.1** The web interface shall be accessible via modern web browsers.
- **NFR-5.2** The system shall provide responsive design supporting desktop, tablet, and mobile devices.
- **NFR-5.3** The system shall provide contextual help.
- **NFR-5.4** 70% of users should be able to complete their desired tasks successfully on the first attempt.
- **NFR-5.5** Users should be able to learn and navigate the system comfortably within 15 minutes of first use.

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6. Integration Requirements

- **NFR-6.1** The system shall support standard data formats (JSON, XML, CSV).
- **NFR-6.2** The system shall implement rate limiting for API calls to prevent abuse.
- **NFR-6.3** The system shall maintain API versioning for backward compatibility.
- **NFR-6.4** The system shall provide webhook functionality for real-time event notifications.

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7. Maintenance Requirements

- **NFR-7.1** The system shall support hot deployment for critical bug fixes.
- **NFR-7.2** The system shall provide comprehensive logging for troubleshooting.
- **NFR-7.3** The system shall include health check endpoints for monitoring.